Chapter 14
Nested and Split-Plot Designs
Solutions

In this chapter we have not shown residual plots and other diagnostics to conserve space. A complete analysis would, of course, include these model adequacy checking procedures.

14.1S. A rocket propellant manufacturer is studying the burning rate of propellant from three production processes. Four batches of propellant are randomly selected from the output of each process and three determinations of burning rate are made on each batch. The results follow. Analyze the data and draw conclusions.

	Process 1				Process 2				Process 3			
Batch	1	2	3	4	1	2	3	4	1	2	3	4
	25	19	15	15	19	23	18	35	14	35	38	25
	30	28	17	16	17	24	21	27	15	21	54	29
	26	20	14	13	14	21	17	25	20	24	50	33

Minitab Output

ANOVA: Burn Rate versus Process, Batch

```
Factor          Type Levels Values
Process        fixed     3    1    2    3
Batch(Process) random     4    1    2    3    4

Analysis of Variance for Burn Rat

Source           DF        SS        MS      F      P
Process           2    676.06    338.03   1.46  0.281
Batch(Process)    9   2077.58    230.84  12.20  0.000
Error            24    454.00     18.92
Total            35   3207.64

Source           Variance Error Expected Mean Square for Each Term
                component term (using restricted model)
 1 Process                2   (3) + 3(2) + 12Q[1]
 2 Batch(Process)    70.64   3   (3) + 3(2)
 3 Error            18.92       (3)
```

There is no significant effect on mean burning rate among the different processes; however, different batches from the same process have significantly different burning rates.

14.2S. A manufacturing engineer is studying the dimensional variability of a particular component that is produced on three machines. Each machine has two spindles, and four components are randomly selected from each spindle. The results follow. Analyze the data, assuming that machines and spindles are fixed factors.

	Machine 1		Machine 2		Machine 3	
Spindle	1	2	1	2	1	2
	12	8	14	12	14	16
	9	9	15	10	10	15
	11	10	13	11	12	15
	12	8	14	13	11	14

Minitab Output

ANOVA: Variability versus Machine, Spindle

```
Factor           Type Levels Values
Machine          fixed     3    1    2    3
Spindle(Machine)  fixed     2    1    2

Analysis of Variance for Variabil

Source            DF        SS        MS      F      P
Machine            2    55.750    27.875   18.93  0.000
Spindle(Machine)     3    43.750    14.583    9.91  0.000
Error             18    26.500     1.472
Total             23   126.000
```

There is a significant effect on dimensional variability due to the machine and spindle factors.

14.3. To simplify production scheduling, an industrial engineer is studying the possibility of assigning one time standard to a particular class of jobs, believing that differences between jobs is negligible. To see if this simplification is possible, six jobs are randomly selected. Each job is given to a different group of three operators. Each operator completes the job twice at different times during the week, and the following results are obtained. What are your conclusions about the use of a common time standard for all jobs in this class? What value would you use for the standard?

	Operator 1		Operator 2		Operator 3	
Job	1	2	1	2	1	2
1	158.3	159.4	159.2	159.6	158.9	157.8
2	154.6	154.9	157.7	156.8	154.8	156.3
3	162.5	162.6	161.0	158.9	160.5	159.5
4	160.0	158.7	157.5	158.9	161.1	158.5
5	156.3	158.1	158.3	156.9	157.7	156.9
6	163.7	161.0	162.3	160.3	162.6	161.8

Minitab Output

ANOVA: Time versus Job, Operator

```
Factor         Type Levels Values
Job          random     6    1    2    3    4    5    6
Operator(Job) random     3    1    2    3

Analysis of Variance for Time

Source          DF        SS        MS      F      P
Job              5   148.111    29.622   17.21  0.000
Operator(Job)    12    20.653     1.721    1.58  0.186
Error           18    19.665     1.092
Total           35   188.430

Source          Variance Error Expected Mean Square for Each Term
             component term (using restricted model)
 1 Job           4.6502   2   (3) + 2(2) + 6(1)
 2 Operator(Job)   0.3143   3   (3) + 2(2)
 3 Error          1.0925      (3)
```

The jobs differ significantly; the use of a common time standard would likely not be a good idea.

14.4S. Consider the three-stage nested design shown in Figure 14.5 to investigate alloy hardness. Using the data that follow, analyze the design, assuming that alloy chemistry and heats are fixed factors and ingots are random. Use the restricted form of the mixed model.

	Alloy Chemistry											
	1						2					
Heats	1		2		3		1		2		3	
Ingots	1	2	1	2	1	2	1	2	1	2	1	2
	40	27	95	69	65	78	22	23	83	75	61	35
	63	30	67	47	54	45	10	39	62	64	77	42

Minitab Output

ANOVA: Hardness versus Alloy, Heat, Ingot

```
Factor            Type Levels Values
Alloy             fixed     2    1    2
Heat(Alloy)       fixed     3    1    2    3
Ingot(Alloy Heat) random    2    1    2

Analysis of Variance for Hardness

Source             DF        SS        MS      F      P
Alloy               1     315.4     315.4   0.85  0.392
Heat(Alloy)         4    6453.8    1613.5   4.35  0.055
Ingot(Alloy Heat)   6    2226.3     371.0   2.08  0.132
Error              12    2141.5     178.5
Total              23   11137.0

Source             Variance Error Expected Mean Square for Each Term
                  component term (using restricted model)
 1 Alloy                      3   (4) + 2(3) + 12Q[1]
 2 Heat(Alloy)                3   (4) + 2(3) + 4Q[2]
 3 Ingot(Alloy Heat)    96.29  4   (4) + 2(3)
 4 Error               178.46      (4)
```

Alloy hardness differs significantly due to the different heats within each alloy.

14.5. Reanalyze the experiment in Problem 14.4 using the unrestricted form of the mixed model. Comment on any differences you observe between the restricted and unrestricted model results. You may use a computer software package.

Minitab Output

ANOVA: Hardness versus Alloy, Heat, Ingot

```
Factor            Type Levels Values
Alloy             fixed     2    1    2
Heat(Alloy)       fixed     3    1    2    3
Ingot(Alloy Heat) random    2    1    2

Analysis of Variance for Hardness

Source             DF        SS        MS      F      P
Alloy               1     315.4     315.4   0.85  0.392
Heat(Alloy)         4    6453.8    1613.5   4.35  0.055
Ingot(Alloy Heat)   6    2226.3     371.0   2.08  0.132
Error              12    2141.5     178.5
Total              23   11137.0

Source             Variance Error Expected Mean Square for Each Term
                  component term (using unrestricted model)
 1 Alloy                      3   (4) + 2(3) + Q[1,2]
 2 Heat(Alloy)                3   (4) + 2(3) + Q[2]
 3 Ingot(Alloy Heat)    96.29  4   (4) + 2(3)
 4 Error               178.46      (4)
```

14.7. Derive the expected means squares for a balanced three-stage nested design, assuming that *A* is fixed and that *B* and *C* are random. Obtain formulas for estimating the variance components.

The expected mean squares can be generated in Minitab as follows:

Minitab Output

ANOVA: y versus A, B, C

```
Factor     Type Levels Values
A         fixed      2    -1    1
B(A)     random      2    -1    1
C(A B)   random      2    -1    1

Analysis of Variance for y

Source      DF        SS        MS      F      P
A            1     0.250     0.250   0.06  0.831
B(A)         2     8.500     4.250   0.35  0.726
C(A B)       4    49.000    12.250   2.13  0.168
Error        8    46.000     5.750
Total       15   103.750

Source      Variance Error Expected Mean Square for Each Term
           component term (using restricted model)
 1 A                2   (4) + 2(3) + 4(2) + 8Q[1]
 2 B(A)      -2.000   3   (4) + 2(3) + 4(2)
 3 C(A B)     3.250   4   (4) + 2(3)
 4 Error      5.750       (4)
```

14.7. Derive the expected means squares for a balanced three-stage nested design if all three factors are random. Obtain formulas for estimating the variance components. Assume the restricted form of the mixed model.

The expected mean squares can be generated in Minitab as follows:

Minitab Output

ANOVA: y versus A, B, C

```
Factor     Type Levels Values
A        random      2    -1    1
B(A)     random      2    -1    1
C(A B)   random      2    -1    1

Analysis of Variance for y

Source      DF        SS        MS      F      P
A            1     0.250     0.250   0.06  0.831
B(A)         2     8.500     4.250   0.35  0.726
C(A B)       4    49.000    12.250   2.13  0.168
Error        8    46.000     5.750
Total       15   103.750

Source      Variance Error Expected Mean Square for Each Term
           component term (using unrestricted model)
 1 A        -0.5000   2   (4) + 2(3) + 4(2) + 8(1)
 2 B(A)     -2.0000   3   (4) + 2(3) + 4(2)
 3 C(A B)    3.2500   4   (4) + 2(3)
 4 Error     5.7500       (4)
```

14.8. Verify the expected mean squares given in Table 14.1.

Factor	F *a* *i*	F *b* *j*	R *n* *l*	E(*MS*)
τ_i	0	b	n	$\sigma^2 + \frac{bn}{a-1}\sum \tau_i^2$
$\beta_{j(i)}$	1	0	n	$\sigma^2 + \frac{n}{a(b-1)}\sum\sum \beta_{j(i)}^2$
$\varepsilon_{(ijk)l}$	1	1	1	σ^2

Factor	R *a* *i*	R *b* *j*	R *n* *l*	E(*MS*)
τ_i	1	b	n	$\sigma^2 + n\sigma_\beta^2 + bn\sigma_\tau^2$
$\beta_{j(i)}$	1	1	n	$\sigma^2 + n\sigma_\beta^2$
$\varepsilon_{(ijk)l}$	1	1	1	σ^2

Factor	F *a* *i*	R *b* *j*	R *n* *l*	E(*MS*)
τ_i	0	b	n	$\sigma^2 + n\sigma_\beta^2 + \frac{bn}{a-1}\sum \tau_i^2$
$\beta_{j(i)}$	1	1	n	$\sigma^2 + n\sigma_\beta^2$
$\varepsilon_{(ijk)l}$	1	1	1	σ^2

14.9. ***Unbalanced nested designs.*** Consider an unbalanced two-stage nested design with b_j levels of B under the ith level of A and n_{ij} replicates in the ijth cell.

(a) Write down the least squares normal equations for this situation. Solve the normal equations.

The least squares normal equations are:

$$\mu = n_{..}\hat{\mu} + \sum_{i=1}^{a} n_{i.}\tau_i + \sum_{i=1}^{a}\sum_{j=1}^{b_i} n_{ij}\hat{\beta}_{j(i)} = y_{...}$$

$$\tau_i = n_{i.}\hat{\mu} + n_{i.}\tau_i + \sum_{j=1}^{b_i} n_{ij}\hat{\beta}_{j(i)} = y_{i..}, \text{ for } i = 1,2,...,a$$

$$\beta_{j(i)} = n_{ij}\hat{\mu} + n_{ij}\tau_i + n_{ij}\hat{\beta}_{j(i)} = y_{ij.}, \text{ for } i = 1,2,...,a \text{ and } j = 1,2,...,b_i$$

There are $1+a+b$ equations in $1+a+b$ unknowns. However, there are $a+1$ linear dependencies in these equations, and consequently, $a+1$ side conditions are needed to solve them. Any convenient set of $a+1$ linearly independent equations can be used. The easiest set is $\hat{\mu} = 0$, $\hat{\tau}_i = 0$, for i=1,2,…,a. Using these conditions we get

$$\hat{\mu} = 0,\ \hat{\tau}_i = 0, \hat{\beta}_{j(i)} = \bar{y}_{ij.}$$

as the solution to the normal equations. See Searle (1971) for a full discussion.

(b) Construct the analysis of variance table for the unbalanced two-stage nested design.

The analysis of variance table is

Source	SS	DF
A	$\sum_{i=1}^{a}\frac{y_{i..}^2}{n_{i.}}-\frac{y_{...}^2}{n_{..}}$	*a*-1
B	$\sum_{i=1}^{a}\sum_{j=1}^{b_i}\frac{y_{ij.}^2}{n_{ij}}-\sum_{i-1}^{a}\frac{y_{i..}^2}{n_{i.}}$	*b.*-*a*
Error	$\sum_{i=1}^{a}\sum_{j=1}^{b_i}\sum_{k=1}^{n_{ij}}y_{ijk}^2-\sum_{i=1}^{a}\sum_{j=1}^{b_i}\frac{y_{ij.}^2}{n_{ij}}$	*n..*-*b*
Total	$\sum_{i=1}^{a}\sum_{j=1}^{b_i}\sum_{k=1}^{n_{ij}}y_{ijk}^2-\frac{y_{...}^2}{n_{..}}$	*n..*-1

(c) Analyze the following data, using the results in part (b).

Factor *A*	1		2		
Factor *B*	1	2	1	2	3
	6	3	5	2	1
	4	1	7	4	0
	8		9	3	-3
			6		

Note that a=2, b_1=2, b_2=3, $b.$=b_1+b_2=5, n_{11}=3, n_{12}=2, n_{21}=4, n_{22}=3 and n_{23}=3

Source	SS	DF	MS
A	0.13	1	0.13
B	153.78	3	51.26
Error	35.42	10	3.54
Total	189.33	14	

The analysis can also be performed in Minitab as follows. The adjusted sum of squares is utilized by Minitab's general linear model routine.

Minitab Output

General Linear Model: y versus A, B

```
Factor    Type Levels Values
A        fixed     2 1 2
B(A)     fixed     5 1 2 1 2 3

Analysis of Variance for y, using Adjusted SS for Tests

Source    DF    Seq SS    Adj SS    Adj MS      F     P
A          1     0.133     0.898     0.898   0.25  0.625
B(A)       3   153.783   153.783    51.261  14.47  0.001
Error     10    35.417    35.417     3.542
Total     14   189.333
```

14.10. A process engineer is testing the yield of a product manufactured on three machines. Each machine can be operated at two power settings. Furthermore, a machine has three stations on which the product is formed. An experiment is conducted in which each machine is tested at both power settings, and three observations on yield are taken from each station. The runs are made in random order, and the results are shown in Table P14.1. Analyze this experiment, assuming all three factors are fixed.

Table P14.1

	Machine 1			Machine 2			Machine 3		
Station	1	2	3	1	2	3	1	2	3
Power	34.1	33.7	36.2	31.1	33.1	32.8	32.9	33.8	33.6
Setting 1	30.3	34.9	36.8	33.5	34.7	35.1	33.0	33.4	32.8
	31.6	35.0	37.1	34.0	33.9	34.3	33.1	32.8	31.7
Power	24.3	28.1	25.7	24.1	24.1	26.0	24.2	23.2	24.7
Setting 2	26.3	29.3	26.1	25.0	25.1	27.1	26.1	27.4	22.0
	27.1	28.6	24.9	26.3	27.9	23.9	25.3	28.0	24.8

The linear model is $y_{ijkl} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \gamma_{k(j)} + (\tau\gamma)_{ik(j)} + \varepsilon_{(ijk)l}$

Minitab Output

ANOVA: Yield versus Machine, Power, Station

```
Factor           Type Levels Values
Machine          fixed     3    1    2    3
Power            fixed     2    1    2
Station(Machine)  fixed     3    1    2    3

Analysis of Variance for Yield

Source                  DF        SS        MS      F     P
Machine                  2    21.436    10.718   6.25  0.005
Power                    1   845.698   845.698 492.96  0.000
Station(Machine)         6    33.583     5.597   3.26  0.012
Machine*Power            2     0.383     0.191   0.11  0.895
Power*Station(Machine)     6    29.208     4.868   2.84  0.023
Error                   36    61.760     1.716
Total                   53   992.068

Source                  Variance Error Expected Mean Square for Each Term
                       component term (using restricted model)
 1 Machine                          6   (6) + 18Q[1]
 2 Power                            6   (6) + 27Q[2]
 3 Station(Machine)                   6   (6) + 6Q[3]
 4 Machine*Power                    6   (6) + 9Q[4]
 5 Power*Station(Machine)             6   (6) + 3Q[5]
 6 Error                  1.716       (6)
```

14.11. Suppose that in Problem 14.10 a large number of power settings could have been used and that the two selected for the experiment were chosen randomly. Obtain the expected mean squares for this situation assuming the restricted form of the mixed model and modify the previous analysis appropriately.

The analysis of variance and the expected mean squares can be obtained from Minitab as follows:

Minitab Output

ANOVA: Yield versus Machine, Power, Station

```
Factor           Type Levels Values
Machine          fixed     3    1    2    3
Power          random      2    1    2
Station(Machine)  fixed     3    1    2    3

Analysis of Variance for Yield

Source                  DF        SS        MS      F     P
Machine                  2    21.436    10.718   56.03  0.018
Power                    1   845.698   845.698 492.96  0.000
Station(Machine)         6    33.583     5.597    1.15  0.435
Machine*Power            2     0.383     0.191    0.11  0.895
Power*Station(Machine)     6    29.208     4.868    2.84  0.023
```

```
Error                  36    61.760     1.716
Total                  53   992.068

Source                  Variance Error Expected Mean Square for Each Term
                      component term (using restricted model)
 1 Machine                          4   (6) + 9(4) + 18Q[1]
 2 Power                 31.2586    6   (6) + 27(2)
 3 Station(Machine)                 5   (6) + 3(5) + 6Q[3]
 4 Machine*Power          -0.1694   6   (6) + 9(4)
 5 Power*Station(Machine)  1.0508   6   (6) + 3(5)
 6 Error                   1.7156       (6)
```

14.12. Reanalyze the experiment in Problem 14.11 assuming the unrestricted form of the mixed model. You may use a computer software program to do this. Comment on any differences between the restricted and unrestricted model analysis and conclusions.

Minitab Output

ANOVA: Yield versus Machine, Power, Station

```
Factor           Type Levels Values
Machine         fixed     3    1    2    3
Power          random     2    1    2
Station(Machine)  fixed     3    1    2    3

Analysis of Variance for Yield

Source                 DF        SS        MS      F     P
Machine                 2    21.436    10.718   56.03  0.018
Power                   1   845.698   845.698 4420.88  0.000
Station(Machine)        6    33.583     5.597    1.15  0.435
Machine*Power           2     0.383     0.191    0.04  0.962
Power*Station(Machine)     6    29.208     4.868    2.84  0.023
Error                  36    61.760     1.716
Total                  53   992.068

Source                  Variance Error Expected Mean Square for Each Term
                      component term (using unrestricted model)
 1 Machine                          4   (6) + 3(5) + 9(4) + Q[1,3]
 2 Power                 31.3151    4   (6) + 3(5) + 9(4) + 27(2)
 3 Station(Machine)                 5   (6) + 3(5) + Q[3]
 4 Machine*Power          -0.5196   5   (6) + 3(5) + 9(4)
 5 Power*Station(Machine)  1.0508   6   (6) + 3(5)
 6 Error                   1.7156       (6)
```

There are differences between several of the expected mean squares. However, the conclusions that could be drawn do not differ in any meaningful way from the restricted model analysis.

14.13. A structural engineer is studying the strength of aluminum alloy purchased from three vendors. Each vendor submits the alloy in standard-sized bars of 1.0, 1.5, or 2.0 inches. The processing of different sizes of bar stock from a common ingot involves different forging techniques, and so this factor may be important. Furthermore, the bar stock if forged from ingots made in different heats. Each vendor submits two tests specimens of each size bar stock from the three heats. The resulting strength data is shown in Table P14.2. Analyze the data, assuming that vendors and bar size are fixed and heats are random. Use the restricted form of the mixed model.

Table P14.2

		Vendor 1			Vendor 2			Vendor 3		
	Heat	1	2	3	1	2	3	1	2	3
Bar	1 inch	1.230	1.346	1.235	1.301	1.346	1.315	1.247	1.275	1.324
Size		1.259	1.400	1.206	1.263	1.392	1.320	1.296	1.268	1.315
	1 ½	1.316	1.329	1.250	1.274	1.384	1.346	1.273	1.260	1.392

inch	1.300	1.362	1.239	1.268	1.375	1.357	1.264	1.265	1.364
2 inch	1.287	1.346	1.273	1.247	1.362	1.336	1.301	1.280	1.319
	1.292	1.382	1.215	1.215	1.328	1.342	1.262	1.271	1.323

$$y_{ijkl} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \gamma_{k(j)} + (\tau\gamma)_{ik(j)} + \varepsilon_{(ijk)l}$$

Minitab Output

ANOVA: Strength versus Vendor, Bar Size, Heat

```
Factor        Type Levels Values
Vendor        fixed     3    1    2    3
Heat(Vendor)  random      3    1    2    3
Bar Size      fixed     3   1.0   1.5   2.0

Analysis of Variance for Strength

Source                 DF        SS        MS      F     P
Vendor                  2  0.0088486  0.0044243   0.26  0.776
Heat(Vendor)            6  0.1002093  0.0167016  41.32  0.000
Bar Size                2  0.0025263  0.0012631   1.37  0.290
Vendor*Bar Size         4  0.0023754  0.0005939   0.65  0.640
Bar Size*Heat(Vendor)  12  0.0110303  0.0009192   2.27  0.037
Error                  27  0.0109135  0.0004042
Total                  53  0.1359034

Source                 Variance Error Expected Mean Square for Each Term
                      component term (using restricted model)
 1 Vendor                        2   (6) + 6(2) + 18Q[1]
 2 Heat(Vendor)          0.00272   6   (6) + 6(2)
 3 Bar Size                      5   (6) + 2(5) + 18Q[3]
 4 Vendor*Bar Size               5   (6) + 2(5) + 6Q[4]
 5 Bar Size*Heat(Vendor)  0.00026   6   (6) + 2(5)
 6 Error                 0.00040      (6)
```

14.14. Rework Problem 14.13 using the unrestricted form of the mixed model. You can use a computer software program to do this. Comment on any differences between the restricted and unrestricted model analysis and conclusions.

Minitab Output

ANOVA: Strength versus Vendor, Bar Size, Heat

```
Factor        Type Levels Values
Vendor        fixed     3    1    2    3
Heat(Vendor)  random      3    1    2    3
Bar Size      fixed     3   1.0   1.5   2.0

Analysis of Variance for Strength

Source                 DF        SS        MS      F     P
Vendor                  2  0.0088486  0.0044243   0.26  0.776
Heat(Vendor)            6  0.1002093  0.0167016  18.17  0.000
Bar Size                2  0.0025263  0.0012631   1.37  0.290
Vendor*Bar Size         4  0.0023754  0.0005939   0.65  0.640
Bar Size*Heat(Vendor)  12  0.0110303  0.0009192   2.27  0.037
Error                  27  0.0109135  0.0004042
Total                  53  0.1359034

Source                 Variance Error Expected Mean Square for Each Term
                      component term (using unrestricted model)
 1 Vendor                        2   (6) + 2(5) + 6(2) + Q[1,4]
 2 Heat(Vendor)          0.00263   5   (6) + 2(5) + 6(2)
 3 Bar Size                      5   (6) + 2(5) + Q[3,4]
 4 Vendor*Bar Size               5   (6) + 2(5) + Q[4]
 5 Bar Size*Heat(Vendor)  0.00026   6   (6) + 2(5)
 6 Error                 0.00040      (6)
```

There are some differences in the expected mean squares. However, the conclusions do not differ from those of the restricted model analysis.

14.15S. Steel in normalized by heating above the critical temperature, soaking, and then air cooling. This process increases the strength of the steel, refines the grain, and homogenizes the structure. An experiment is performed to determine the effect of temperature and heat treatment time on the strength of normalized steel. Two temperatures and three times are selected. The experiment is performed by heating the oven to a randomly selected temperature and inserting three specimens. After 10 minutes one specimen is removed, after 20 minutes the second specimen is removed, and after 30 minutes the final specimen is removed. Then the temperature is changed to the other level and the process is repeated. Four shifts are required to collect the data, which are shown below. Analyze the data and draw conclusions, assume both factors are fixed.

		Temperature (F)	
Shift	Time(minutes)	1500	1600
1	10	63	89
	20	54	91
	30	61	62
2	10	50	80
	20	52	72
	30	59	69
3	10	48	73
	20	74	81
	30	71	69
4	10	54	88
	20	48	92
	30	59	64

This is a split-plot design. Shifts correspond to blocks, temperature is the whole plot treatment, and time is the sub-treatments (in the subplot or split-plot part of the design). The expected mean squares and analysis of variance are shown below. The following Minitab Output has been modified to display the results of the split-plot analysis. Minitab will calculate the sums of squares correctly, but the expected mean squares and the statistical tests are not, in general, correct. Notice that the Error term in the analysis of variance is actually the three factor interaction.

Minitab Output

ANOVA: Strength versus Shift, Temperature, Time

```
Factor    Type Levels Values
Shift   random     4    1    2    3    4
Temperat  fixed      2  1500  1600
Time      fixed      3   10   20   30

Analysis of Variance for Strength
                                          Standard    Split Plot
Source          DF       SS       MS      F     P      F     P
Shift            3    145.46    48.49   1.19  0.390
Temperat         1   2340.38  2340.38  29.20  0.012  29.21  0.012
Shift*Temperat   3    240.46    80.15   1.97  0.220
Time             2    159.25    79.63   1.00  0.422   1.00  0.422
Shift*Time       6    478.42    79.74   1.96  0.217
Temperat*Time    2    795.25   397.63   9.76  0.013   9.76  0.013
Error            6    244.42    40.74
Total           23   4403.63

Source          Variance Error Expected Mean Square for Each Term
               component term (using restricted model)
 1 Shift           1.292   7   (7) + 6(1)
```

```
 2 Temperat               3   (7) + 3(3) + 12Q[2]
 3 Shift*Temperat   13.139   7   (7) + 3(3)
 4 Time                   5   (7) + 2(5) + 8Q[4]
 5 Shift*Time      19.500   7   (7) + 2(5)
 6 Temperat*Time           7   (7) + 4Q[6]
 7 Error          40.736       (7)
```

14.16S. An experiment is designed to study pigment dispersion in paint. Four different mixes of a particular pigment are studied. The procedure consists of preparing a particular mix and then applying that mix to a panel by three application methods (brushing, spraying, and rolling). The response measured is the percentage reflectance of the pigment. Three days are required to run the experiment, and the data obtained follow. Analyze the data and draw conclusions, assuming that mixes and application methods are fixed.

		Mix			
Day	App Method	1	2	3	4
1	1	64.5	66.3	74.1	66.5
	2	68.3	69.5	73.8	70.0
	3	70.3	73.1	78.0	72.3
2	1	65.2	65.0	73.8	64.8
	2	69.2	70.3	74.5	68.3
	3	71.2	72.8	79.1	71.5
3	1	66.2	66.5	72.3	67.7
	2	69.0	69.0	75.4	68.6
	3	70.8	74.2	80.1	72.4

This is a split plot design. Days correspond to blocks, mix is the whole plot treatment, and method is the sub-treatment (in the subplot or split plot part of the design). The following Minitab Output has been modified to display the results of the split-plot analysis. Minitab will calculate the sums of squares correctly, but the expected mean squares and the statistical tests are not, in general, correct. Notice that the Error term in the analysis of variance is actually the three factor interaction.

Minitab Output

ANOVA: Reflectance versus Day, Mix, Method

```
Factor    Type Levels Values
Day     random     3    1    2    3
Mix      fixed     4    1    2    3    4
Method   fixed     3    1    2    3

Analysis of Variance for Reflecta
                                        Standard    Split Plot
Source       DF        SS        MS      F     P      F     P
Day           2     2.042     1.021   1.39  0.285
Mix           3   307.479   102.493  135.77  0. 000 135.75  0.000
Day*Mix       6     4.529     0.755   1.03  0.451
Method        2   222.095   111.047  226.24  0.000  226.16  0.000
Day*Method    4     1.963     0.491   0.67  0.625
Mix*Method    6    10.036     1.673   2.28  0.105   2.28  0.105
Error        12     8.786     0.732
Total        35   556.930

Source       Variance Error Expected Mean Square for Each Term
          component term (using restricted model)
 1 Day        0.02406   7   (7) + 12(1)
 2 Mix                3   (7) + 3(3) + 9Q[2]
 3 Day*Mix    0.00759   7   (7) + 3(3)
 4 Method             5   (7) + 4(5) + 12Q[4]
 5 Day*Method -0.06032   7   (7) + 4(5)
 6 Mix*Method         7   (7) + 3Q[6]
 7 Error      0.73213      (7)
```

14.17. Repeat Problem 14.16, assuming that the mixes are random and the application methods are fixed.

The F-tests are the same as those in Problem 14-16. The following Minitab Output has been edited to display the results of the split-plot analysis. Minitab will calculate the sums of squares correctly, but the expected mean squares and the statistical tests are not, in general, correct. Again, the Error term in the analysis of variance is actually the three factor interaction.

Minitab Output

ANOVA: Reflectance versus Day, Mix, Method

```
Factor    Type Levels Values
Day     random      3    1    2    3
Mix     random      4    1    2    3    4
Method   fixed      3    1    2    3

Analysis of Variance for Reflecta
                                        Standard       Split Plot
Source       DF        SS        MS       F     P         F     P
Day           2     2.042     1.021    1.35  0.328
Mix           3   307.479   102.493  135.77  0.000    135.75  0.000
Day*Mix       6     4.529     0.755    1.03  0.451
Method        2   222.095   111.047   77.58  0.001 x   226.16  0.000
Day*Method    4     1.963     0.491    0.67  0.625
Mix*Method    6    10.036     1.673    2.28  0.105      2.28  0.105
Error        12     8.786     0.732
Total        35   556.930

x Not an exact F-test.

Source       Variance Error Expected Mean Square for Each Term
          component term (using restricted model)
 1 Day         0.0222   3   (7) + 3(3) + 12(1)
 2 Mix        11.3042   3   (7) + 3(3) + 9(2)
 3 Day*Mix     0.0076   7   (7) + 3(3)
 4 Method             *   (7) + 3(6) + 4(5) + 12Q[4]
 5 Day*Method -0.0603   7   (7) + 4(5)
 6 Mix*Method  0.3135   7   (7) + 3(6)
 7 Error       0.7321       (7)

* Synthesized Test.

Error Terms for Synthesized Tests

Source        Error DF  Error MS  Synthesis of Error MS
 4 Method         3.59     1.431   (5) + (6) - (7)
```

14.18. Consider the split-split-plot design described in Example 14.4. Suppose that this experiment is conducted as described and that the data shown in Table P14.3 are obtained. Analyze the data and draw conclusions.

Table P14.3

		Technician								
		1			2			3		
Blocks	Dose Strength	1	2	3	1	2	3	1	2	3
	Wall Thickness									
1	1	95	71	108	96	70	108	95	70	100
	2	104	82	115	99	84	100	102	81	106
	3	101	85	117	95	83	105	105	84	113
	4	108	85	116	97	85	109	107	87	115
2	1	95	78	110	100	72	104	92	69	101
	2	106	84	109	101	79	102	100	76	104
	3	103	86	116	99	80	108	101	80	109

	4	109	84	110	112	86	109	108	86	113
3	1	96	70	107	94	66	100	90	73	98
	2	105	81	106	100	84	101	97	75	100
	3	106	88	112	104	87	109	100	82	104
	4	113	90	117	121	90	117	110	91	112
4	1	90	68	109	98	68	106	98	72	101
	2	100	84	112	102	81	103	102	78	105
	3	102	85	115	100	85	110	105	80	110
	4	114	88	118	118	85	116	110	95	120

Using the computer output, the F-ratios were calculated by hand using the expected mean squares found in Table 14.25. The following Minitab Output has been edited to display the results of the split-plot analysis. Minitab will calculate the sums of squares correctly, but the expected mean squares and the statistical tests are not, in general, correct. Notice that the Error term in the analysis of variance is actually the four factor interaction.

Minitab Output

ANOVA: Time versus Day, Tech, Dose, Thick

```
Factor    Type Levels Values
Day     random      4    1    2    3    4
Tech     fixed      3    1    2    3
Dose     fixed      3    1    2    3
Thick    fixed      4    1    2    3    4

Analysis of Variance for Time
                                          Standard    Split Plot
Source           DF       SS       MS      F     P      F     P
Day               3    48.41    16.14   3.38  0.029
Tech              2   248.35   124.17   4.62  0.061   4.62  0.061
Day*Tech          6   161.15    26.86   5.62  0.000
Dose              2 20570.06 10285.03 550.44  0.000 550.30  0.000
Day*Dose          6   112.11    18.69   3.91  0.004
Tech*Dose         4   125.94    31.49   3.32  0.048   3.32  0.048
Day*Tech*Dose    12   113.89     9.49   1.99  0.056
Thick             3  3806.91  1268.97  36.47  0.000  36.48  0.000
Day*Thick         9   313.12    34.79   7.28  0.000
Tech*Thick        6   126.49    21.08   2.26  0.084   2.26  0.084
Day*Tech*Thick   18   167.57     9.31   1.95  0.044
Dose*Thick        6   402.28    67.05  17.13  0.000  17.15  0.000
Day*Dose*Thick   18    70.44     3.91   0.82  0.668
Tech*Dose*Thick  12   205.89    17.16   3.59  0.001   3.59  0.001
Error            36   172.06     4.78
Total           143 26644.66

Source           Variance Error Expected Mean Square for Each Term
                component term (using restricted model)
 1 Day             0.3155  15   (15) + 36(1)
 2 Tech                    3   (15) + 12(3) + 48Q[2]
 3 Day*Tech         1.8400  15   (15) + 12(3)
 4 Dose                    5   (15) + 12(5) + 48Q[4]
 5 Day*Dose         1.1588  15   (15) + 12(5)
 6 Tech*Dose                7   (15) + 4(7) + 16Q[6]
 7 Day*Tech*Dose    1.1779  15   (15) + 4(7)
 8 Thick                   9   (15) + 9(9) + 36Q[8]
 9 Day*Thick        3.3346  15   (15) + 9(9)
10 Tech*Thick              11   (15) + 3(11) + 12Q[10]
11 Day*Tech*Thick   1.5100  15   (15) + 3(11)
12 Dose*Thick              13   (15) + 3(13) + 12Q[12]
13 Day*Dose*Thick  -0.2886  15   (15) + 3(13)
14 Tech*Dose*Thick         15   (15) + 4Q[14]
15 Error           4.7793      (15)
```

14.19. Rework Problem 14.18, assuming that the dosage strengths are chosen at random. Use the restricted form of the mixed model.

The following Minitab Output has been edited to display the results of the split-plot analysis. Minitab will calculate the sums of squares correctly, but the expected mean squares and the statistical tests are not, in general, correct. Again, the Error term in the analysis of variance is actually the four factor interaction.

Minitab Output

ANOVA: Time versus Day, Tech, Dose, Thick

```
Factor    Type Levels Values
Day     random     4    1    2    3    4
Tech     fixed     3    1    2    3
Dose    random     3    1    2    3
Thick    fixed     4    1    2    3    4

Analysis of Variance for Time
                                          Standard       Split Plot
Source           DF        SS        MS      F     P         F     P
Day               3     48.41     16.14   0.86  0.509
Tech              2    248.35    124.17   2.54  0.155      4.62  0.061
Day*Tech          6    161.15     26.86   2.83  0.059
Dose              2  20570.06  10285.03 550.44  0.000    550.30  0.000
Day*Dose          6    112.11     18.69   3.91  0.004
Tech*Dose         4    125.94     31.49   3.32  0.048      3.32  0.048
Day*Tech*Dose    12    113.89      9.49   1.99  0.056
Thick             3   3806.91   1268.97  12.96  0.001 x   36.48  0.000
Day*Thick         9    313.12     34.79   8.89  0.000
Tech*Thick        6    126.49     21.08   0.97  0.475 x    2.26  0.084
Day*Tech*Thick   18    167.57      9.31   1.95  0.044
Dose*Thick        6    402.28     67.05  17.13  0.000     17.15  0.000
Day*Dose*Thick   18     70.44      3.91   0.82  0.668
Tech*Dose*Thick  12    205.89     17.16   3.59  0.001      3.59  0.001
Error            36    172.06      4.78
Total           143  26644.66

x Not an exact F-test.

Source           Variance Error Expected Mean Square for Each Term
                component term (using restricted model)
 1 Day             -0.071   5   (15) + 12(5) + 36(1)
 2 Tech                    *   (15) + 4(7) + 16(6) + 12(3) + 48Q[2]
 3 Day*Tech         1.447   7   (15) + 4(7) + 12(3)
 4 Dose           213.882   5   (15) + 12(5) + 48(4)
 5 Day*Dose         1.159  15   (15) + 12(5)
 6 Tech*Dose        1.375   7   (15) + 4(7) + 16(6)
 7 Day*Tech*Dose    1.178  15   (15) + 4(7)
 8 Thick                 *   (15) + 3(13) + 12(12) + 9(9) + 36Q[8]
 9 Day*Thick        3.431  13   (15) + 3(13) + 9(9)
10 Tech*Thick             *   (15) + 4(14) + 3(11) + 12Q[10]
11 Day*Tech*Thick   1.510  15   (15) + 3(11)
12 Dose*Thick       5.261  13   (15) + 3(13) + 12(12)
13 Day*Dose*Thick  -0.289  15   (15) + 3(13)
14 Tech*Dose*Thick  3.095  15   (15) + 4(14)
15 Error           4.779      (15)

* Synthesized Test.

Error Terms for Synthesized Tests

Source           Error DF  Error MS  Synthesis of Error MS
 2 Tech             6.35     48.85   (3) + (6) - (7)
 8 Thick           10.84     97.92   (9) + (12) - (13)
10 Tech*Thick       15.69     21.69   (11) + (14) - (15)
```

There are no exact tests on technicians β_j, dosage strengths γ_k, wall thickness δ_h, or the technician x wall thickness interaction $(\beta\delta)_{jh}$. The approximate *F*-tests are as follows:

$H_0: \beta_j = 0$

$$F = \frac{MS_B + MS_{ABC}}{MS_{AB} + MS_{BC}} = \frac{124.174 + 9.491}{26.859 + 31.486} = 2.291$$

$$p = \frac{\left(MS_B + MS_{ABC}\right)^2}{\frac{MS_B^2}{2} + \frac{MS_{ABC}^2}{12}} = \frac{\left(124.174 + 9.491\right)^2}{\frac{124.174^2}{2} + \frac{9.491^2}{12}} = 2.315$$

$$q = \frac{\left(MS_{AB} + MS_{BC}\right)^2}{\frac{MS_{AB}^2}{6} + \frac{MS_{BC}^2}{4}} = \frac{\left(26.859 + 31.486\right)^2}{\frac{26.859^2}{6} + \frac{31.486^2}{4}} = 9.248$$

Do not reject $H_0: \beta_j = 0$

$H_0: \gamma_k = 0$

$$F = \frac{MS_C + MS_{ACD}}{MS_{CD} + MS_{AD}} = \frac{10285.028 + 3.914}{67.046 + 34.791} = 101.039$$

$$p = \frac{\left(MS_C + MS_{ACD}\right)^2}{\frac{MS_C^2}{2} + \frac{MS_{ACD}^2}{18}} = \frac{\left(10285.028 + 3.914\right)^2}{\frac{10285.028^2}{2} + \frac{3.914^2}{18}} = 2.002$$

$$q = \frac{\left(MS_{CD} + MS_{AD}\right)^2}{\frac{MS_{CD}^2}{6} + \frac{MS_{AD}^2}{9}} = \frac{\left(67.046 + 34.791\right)^2}{\frac{67.046^2}{6} + \frac{34.791^2}{9}} = 11.736$$

Reject $H_0: \gamma_k = 0$

$H_0: \delta_h = 0$

$$F = \frac{MS_D + MS_{ACD}}{MS_{CD} + MS_{AD}} = \frac{1268.970 + 3.914}{67.046 + 34.791} = 12.499$$

$$p = \frac{\left(MS_D + MS_{ACD}\right)^2}{\frac{MS_D^2}{3} + \frac{MS_{ACD}^2}{18}} = \frac{\left(1268.970 + 3.914\right)^2}{\frac{1268.970^2}{3} + \frac{3.914^2}{18}} = 3.019$$

$$q = \frac{\left(MS_{CD} + MS_{AD}\right)^2}{\frac{MS_{CD}^2}{6} + \frac{MS_{AD}^2}{9}} = \frac{\left(67.046 + 34.791\right)^2}{\frac{67.046^2}{6} + \frac{34.791^2}{9}} = 11.736$$

Reject $H_0: \delta_h = 0$

$H_0: \ (\beta\delta)_{jh} = 0$

$$F = \frac{MS_{BD} + MS_{ABCD}}{MS_{BCD} + MS_{ABD}} = \frac{21.081 + 4.779}{17.157 + 9.309} = 0.977$$

$F < 1$, do not reject $H_0: \ (\beta\delta)_{jh} = 0$

14.20. Suppose that in Problem 14.18 four technicians had been used. Assuming that all the factors are fixed, how many blocks should be run to obtain an adequate number of degrees of freedom on the test for differences among technicians?

The number of degrees of freedom for the test is (a-1)(4-1)=3(a-1), where a is the number of blocks used.

Number of Blocks (a)	DF for test
2	3
3	6
4	9
5	12

At least three blocks should be run, but four would give a better test.

14.21. Consider the experiment described in Example 14.4. Demonstrate how the order in which the treatments combinations are run would be determined if this experiment were run as

(a) a split-split-plot

Randomization for the split-split plot design is described in Example 14.4.

(b) a split-plot

In the split-plot, within a block, the technicians would be the main treatment and within a block-technician plot, the 12 combinations of dosage strength and wall thickness would be run in random order. The design would be a two-factor factorial in a split-plot.

(c) a factorial design in a randomized block

To run the design in a randomized block, the 36 combinations of technician, dosage strength, and wall thickness would be run in random order within each block. The design would be a three factor factorial in a randomized block.

(d) a completely randomized factorial design.

The blocks would be considered as replicates, and all 144 observations would be 4 replicates of a three factor factorial.

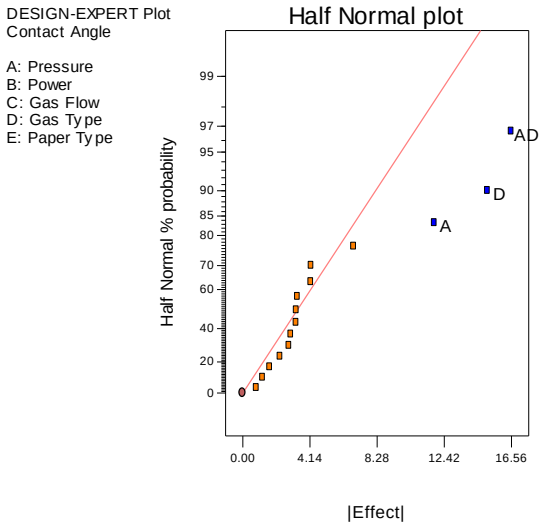
14.22. An article in *Quality Engineering* ("Quality Quandaries: Two-Level Factorials Run as Split-Plot Experiments", Bisgaard, et al, Vol. 8, No. 4, pp. 705-708, 1996) describes a 2^5 factorial experiment on a plasma process focused on making paper more susceptible to ink. Four of the factors (*A*-*D*) are difficult to change from run-to-run, so the experimenters set up the reactor at the eight sets of conditions specified by the low and high levels of these factors, and then processed the two paper types (factor *E*) together. The placement of the paper specimens in the reactors (right versus left) was randomized. This produces a

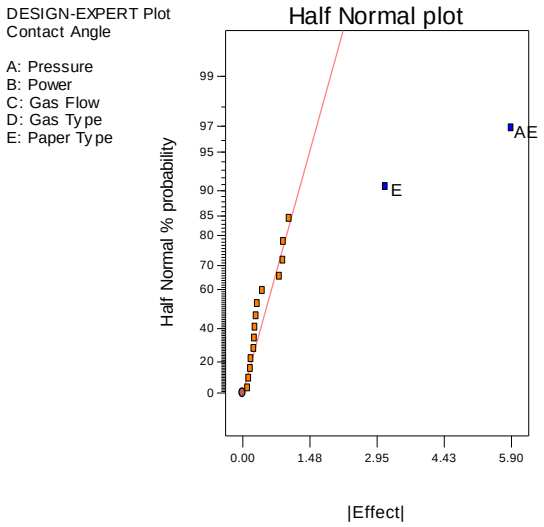
split-plot design with *A*-*D* as the whole-plot factors and factor *E* as the subplot factor. The data from this experiment are shown in Table P14.4. Analyze the data from this experiment and draw conclusions.

Table P14.4

Standard Order	Run Number	A = Pressure	B = Power	C = Gas Flow	D = Gas Type	E = Paper Type	y Contact Angle
1	23	-1	-1	-1	Oxygen	E1	48.6
2	3	+1	-1	-1	Oxygen	E1	41.2
3	11	-1	+1	-1	Oxygen	E1	55.8
4	29	+1	+1	-1	Oxygen	E1	53.5
5	1	-1	-1	+1	Oxygen	E1	37.6
6	15	+1	-1	+1	Oxygen	E1	47.2
7	27	-1	+1	+1	Oxygen	E1	47.2
8	25	+1	+1	+1	Oxygen	E1	48.7
9	19	-1	-1	-1	SiCl4	E1	5.0
10	5	+1	-1	-1	SiCl4	E1	56.8
11	9	-1	+1	-1	SiCl4	E1	25.6
12	31	+1	+1	-1	SiCl4	E1	41.8
13	13	-1	-1	+1	SiCl4	E1	13.3
14	7	+1	-1	+1	SiCl4	E1	47.5
15	21	-1	+1	+1	SiCl4	E1	11.3
16	17	+1	+1	+1	SiCl4	E1	49.5
17	24	-1	-1	-1	Oxygen	E2	57.0
18	4	+1	-1	-1	Oxygen	E2	38.2
19	12	-1	+1	-1	Oxygen	E2	62.9
20	30	+1	+1	-1	Oxygen	E2	51.3
21	2	-1	-1	+1	Oxygen	E2	43.5
22	16	+1	-1	+1	Oxygen	E2	44.8
23	28	-1	+1	+1	Oxygen	E2	54.6
24	26	+1	+1	+1	Oxygen	E2	44.4
25	20	-1	-1	-1	SiCl4	E2	18.1
26	6	+1	-1	-1	SiCl4	E2	56.2
27	10	-1	+1	-1	SiCl4	E2	33.0
28	32	+1	+1	-1	SiCl4	E2	37.8
29	14	-1	-1	+1	SiCl4	E2	23.7
30	8	+1	-1	+1	SiCl4	E2	43.2
31	22	-1	+1	+1	SiCl4	E2	23.9
32	18	+1	+1	+1	SiCl4	E2	48.2

Half normal probability plots of the effects for both the whole plot with factors *A*, *B*, *C*, *D*, and their corresponding interactions, as well as the sub-plot with factor *E* and all interactions involving *E*, are shown below. The analysis of variance is not shown because of the known errors in the calculations; however, the models are also shown below.





Design Expert Output

Response: Contact Angle

Final Equation in Terms of Coded Factors:

Contact Angle =
+40.98
+5.91 * A
-7.55 * D
+1.57 * E
+8.28 * A * D
-2.95 * A * E

Final Equation in Terms of Actual Factors:

Gas Type Oxygen
Paper Type E1
Contact Angle =
+46.96250
+0.58125 * Pressure

Gas Type SiCl4
Paper Type E1
Contact Angle =
+31.86250
+17.14375 * Pressure

Gas Type Oxygen
Paper Type E2
Contact Angle =
+50.10000
-5.31875 * Pressure

Gas Type SiCl4
Paper Type E2
Contact Angle =
+35.00000
+11.24375 * Pressure

14.23. There is always an interaction term in a nested design.

True **False**

14.24. The nested factor is always random.

True **False**

14.25. Both the ANOVA and the REML can be applied to the nested design.

True False

14.26. If there are 3 levels of the main factor and 5 levels of the nested factor there will be 4 degrees of freedom for the nested factor.

True **False**

14.28. A good reason to consider a split-plot design is a situation where some experimental units are larger than others.

True False

14.29. In a split-plot design the subplot error is usually smaller than the whole plot error.

True False

14.30. If an experiment is conducted as a split-plot but analyzed as a factorial, the resulting error estimate will usually be too large for properly testing the subplot factor and could lead to misleading conclusions.

True False

14.31. It is usually cheaper to change the levels of the whole plot factor in a split-plot design.

True **False**

14.32. In and experiment that has at least one nested factor

(a) The nested factor will always be a random effect.
(b) The primary factor will always be a fixed effect.
(c) One may identify the key factor by arbitrarily renumbering the levels.
(d) None of the above (a-c) are true.
(e) All of above (a-c) are true.